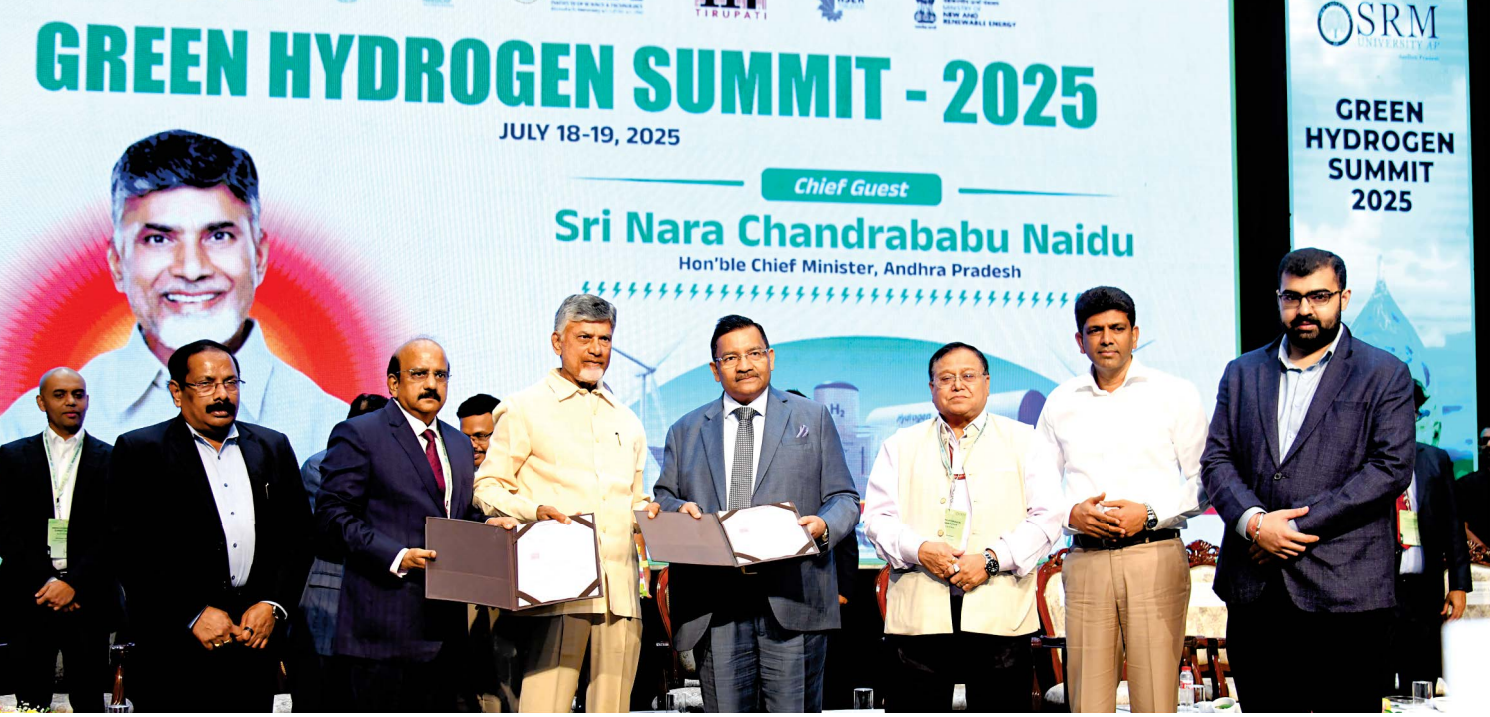


hydrogen and green ammonia before being supplied to domestic consumers or exported through ports.

Each location brings a distinct advantage. Visakhapatnam offers an established industrial base, including refinery and steel demand. Kakinada provides deep-water port infrastructure and experience in handling liquid ammonia. Pudimadaka is emerging as a dedicated green-hydrogen hub.

At Pudimadaka in Anakapalli district, NTPC Green Energy is developing a Green Hydrogen Hub on about 1,200 acres. Current project plans refer to 7 GW of renewable energy and more than 2.5 million tonnes of green chemicals. As of mid-2026, however, the project remained at the enabling-infrastructure stage, with land and site-development activities underway.

Kakinada is further along the value chain. AM Green is developing a green-ammonia project on the former Nagarjuna Fertilisers site spread across approximately 495 acres. Financial closure was achieved in August 2024 and construction began in January 2026. Depending on project documentation, planned capacity ranges from 1.09 MTPA in the initial phases to a broader long-term target of 2 MTPA.

ReNew is developing a project comprising approximately 1.8 GWp of solar capacity, 1 GW of wind generation and 2 GWh of battery storage, backed by a reported investment of around ₹22,000 crore.

These projects reflect both the opportunity and uncertainty surrounding green hydrogen. India has set a national target of producing 5 MTPA of green hydrogen by 2030, yet commercial-scale production remains at an early stage. For Andhra Pradesh, early-mover advantage could be significant, provided projects secure reliable power, transmission, water, storage and long-term buyers.

RENEWABLE POWER NEEDS A BACKBONE

If the coastline is where green molecules could be produced and exported, Rayalaseema is positioning itself as the energy backbone that powers the system.

The Kurnool-Nandyal region hosts Greenko's Pinnapuram integrated renewable-energy and storage project, combining 4,000 MW of solar generation, 1,000 MW of wind capacity and 1,680 MW of pumped-storage capability. According to Ministry of Power statements, all eight pumped-storage units were commissioned during 2025-26.

The significance lies not merely in scale but in reliability. Solar and wind power are inherently variable. Storage allows renewable electricity to be dispatched when required, making it more suitable for industrial consumption and round-the-clock supply commitments.

Anantapur is emerging as another major renewable-energy centre. ReNew is developing a project comprising approximately 1.8 GWp of solar capacity, 1 GW of wind generation and 2 GWh of battery storage, backed by a reported investment of around ₹22,000 crore.

The company is also constructing a ₹4,200-crore solar manufacturing facility in Anakapalli with planned capacity of 6.5 GW for ingots and wafers. The project signals an important shift in Andhra Pradesh's clean-energy strategy: moving beyond power generation and into manufacturing linked to the renewable-energy value chain.

FROM POWER TO PRODUCTS

The larger opportunity for Andhra Pradesh is not simply producing renewable electricity but using that electricity to create higher-value products.

Green hydrogen can support refining, chemicals, fertilisers, steel manufacturing and, potentially, shipping. Green ammonia can become both an industrial feedstock and an export product. Reliable renewable power can attract energy-intensive industries, while storage can improve industrial competitiveness.

The emerging model can be understood as a connected chain:

Renewable Power → Storage → Green Hydrogen → Ammonia & Industrial Products → Ports → Global Markets

This integration is particularly relevant because



